

Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 6:

National Income Accounting

► What you will learn in this chapter

- How economists use aggregate measures to track the performance of the economy
- What gross domestic product, or GDP, is and the three ways of calculating it
- The difference between real GDP and nominal GDP and why real GDP is the appropriate measure of real economic activity
- What a price index is and how it is used to calculate the inflation rate

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Economy Showing Strength in Early 2023 After Last Quarter's GDP Gain Revised Modestly Lower

Output expanded at 2.7% rate in final quarter of 2022, Commerce Department says

How can you compare the sizes of two economies when they produce different things?

By comparing the *value of their production*.

GDP (gross domestic product) is the most important and common way to estimate an economy's size.

THE NATIONAL ACCOUNTS

The national income and product accounts (NIPA)

- measure our nation's economic performance
- compare American income and output to that of other nations
- track the economy's condition throughout the business cycle



THE NATIONAL ACCOUNTS

Consumer spending: household spending on goods and services

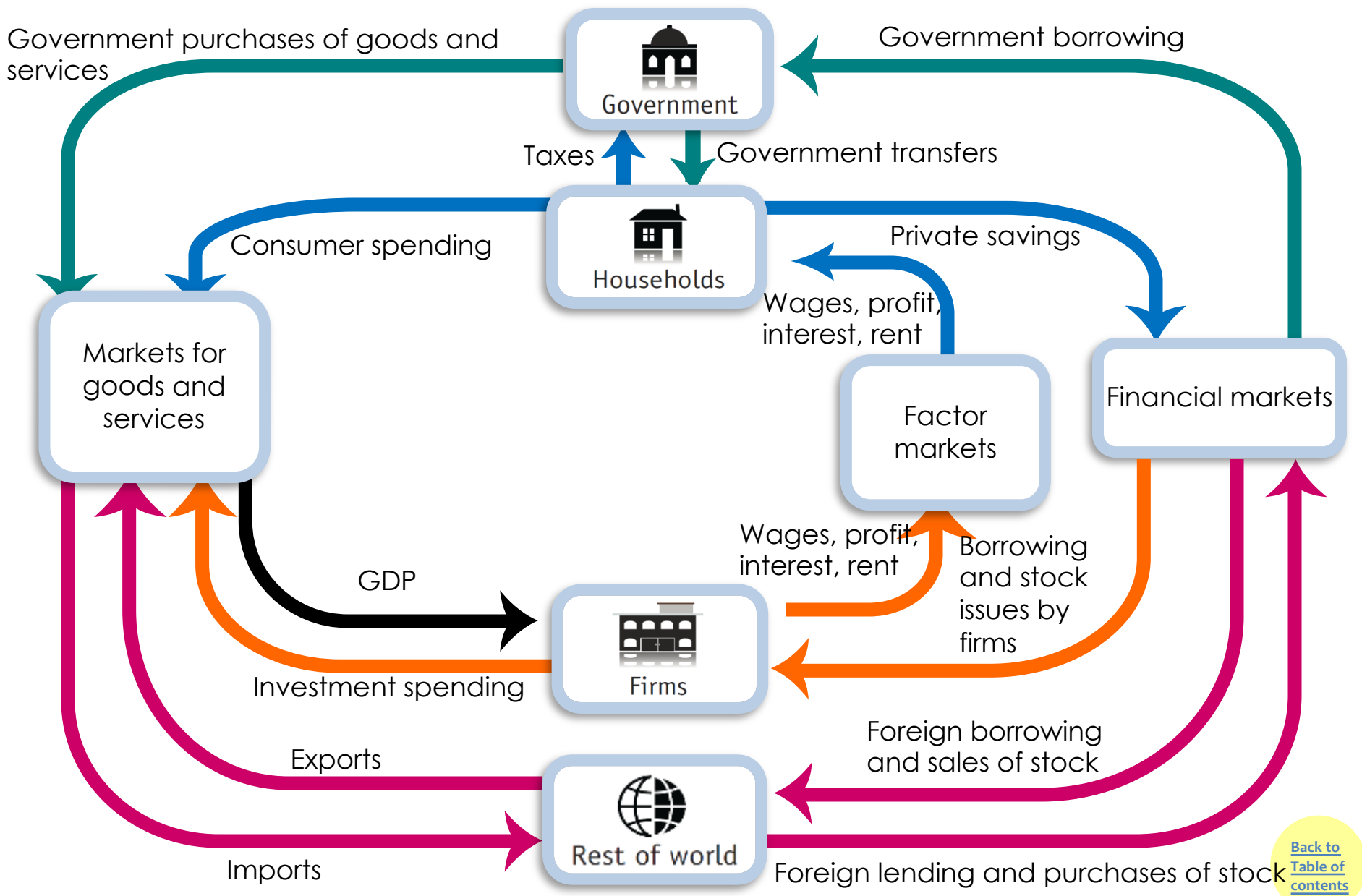
Stock: a share in the ownership of a company held by a shareholder.

Bond: borrowing in the form of an IOU that pays interest.

Government transfer: payment by the government to individuals for which no good or service is provided in return.

Disposable income: income plus government transfers minus taxes; available to spend on consumption and to save.

AN EXPANDED CIRCULAR-FLOW DIAGRAM



THE NATIONAL ACCOUNTS

Households don't spend all of their disposable income. Some of it is saved in the financial markets.

Private savings: disposable income minus consumer spending.

Financial markets: the **banking, stock, and bond markets**, which channel private savings and foreign lending into investment spending, government borrowing, and foreign borrowing.

THE NATIONAL ACCOUNTS

The government spends and borrows for various reasons.

Government borrowing: the total amount of funds borrowed by federal, state, and local governments in the financial markets.

Government purchases of goods and services: total expenditures on goods and services by federal, state, and local governments.

THE NATIONAL ACCOUNTS

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THE NATIONAL ACCOUNTS



Exports: goods and services sold to other countries.

Imports: goods and services purchased from other countries.

THE NATIONAL ACCOUNTS

Inventories: stocks of goods and raw materials held to facilitate business operations.

Investment spending: spending on productive physical capital, such as machinery and construction of structures, and on changes to inventories.

WHAT IS GDP?

Gross domestic product (GDP):
the market value
of all final
goods and services
produced
within a country
in a year.

CALCULATING GROSS DOMESTIC PRODUCT

GDP can be calculated three ways:

1. Add up the *total value of all final goods and services produced*
2. Add up all *spending on domestically produced final goods and services.*
3. Add up the *total factor income earned by households* from firms in the economy

SPENDING = INCOME

It doesn't matter HOW we measure the production, since one person's spending is another's income.



\$10 for a shave? Barber's income of \$10 = customer's spending of \$10.

WHAT IS GDP?

...Produced...

GDP measures *production*.

Sale of *used* goods: **NOT** included.

The sale of *financial assets*, such as stocks and bonds, are not included.



This building's value was counted when it was built.

WHAT IS GDP?

...Within a Country...

Only production that takes place within the borders of a country is included in GDP.

Examples:

**Cars produced in Mexico by American firms:
NOT included in the U.S. GDP.**

**Cars produced in the U.S. by Japanese firms
ARE included in the U.S. GDP.**

LEARN BY DOING: PRACTICE QUESTION

Are the following included in U.S. GDP? (1) The price paid by a German tourist when staying at a New York hotel. (2) The price paid by an American tourist when staying at a Berlin hotel.

- a) Both are included in U.S. GDP.**
- b) Neither is included in U.S. GDP.**
- c) Only the price paid by a German tourist when staying at a New York hotel is included in U.S. GDP.**
- d) Only the price paid by an American tourist when staying at a Berlin hotel is included U.S. GDP.**

WHAT IS GDP?

...in a year.

GDP is like *annual income*:

it measures a rate of production during a given period.

For the current quarter's GDP data, visit the U.S. Bureau of Economic Analysis [here](#).



GDP: WHAT'S IN AND WHAT'S OUT

Included

- Domestically produced final goods and services, including capital goods, new construction of structures, and changes to inventories

Not Included

- Intermediate goods and services
- Inputs
- Used goods
- Financial assets, such as stocks and bonds
- Goods and services produced outside this country



OUR IMPUTED LIVES

Problem: GDP does not count the value of services family members provide to each other... except for the estimated value of housing “services” for those that own their home (instead of renting)

METHOD 1: VALUE ADDED

Value added of a producer is the value of its sales minus the value of its purchases of intermediate goods and services.



+ value added **=**



METHOD 2: SPENDING

Final goods and services: goods and services sold to the final, or end, user.

Intermediate goods and services: goods and services (bought from one firm by another firm) that are **inputs** for production of **final goods and services**.

METHOD 2: SPENDING

Add up all spending on domestically produced final goods and services.

This results in the equation $GDP = C + I + G + X - IM$ where

C = consumer spending,

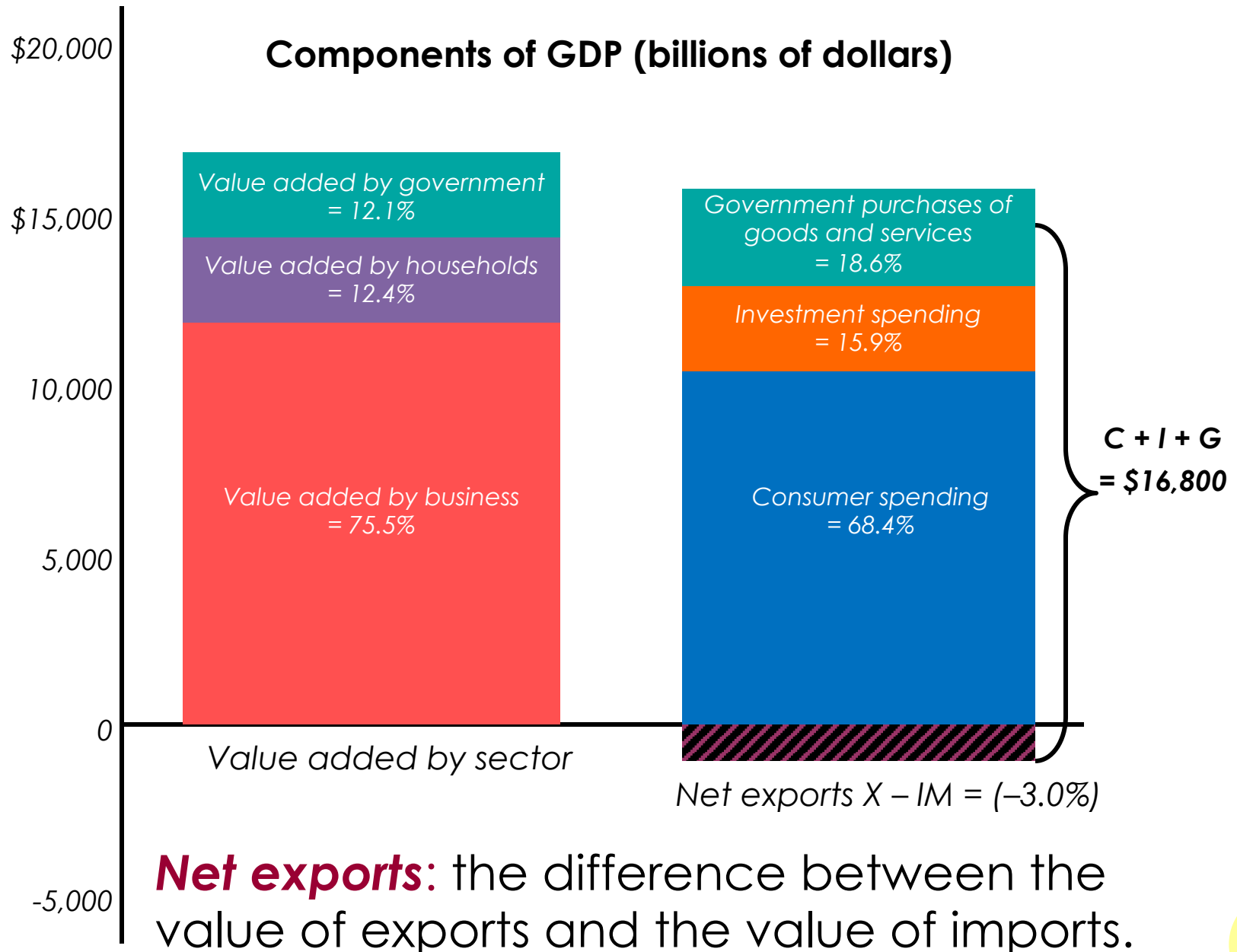
I = investment spending,

G = government purchases of goods and services,

X = sales to foreigners, and

IM = imports (purchases here of foreign goods... or income that has leaked across national borders).

VALUE ADDED VS. SPENDING METHOD



CALCULATING GROSS DOMESTIC PRODUCT

2. Aggregate spending on domestically produced final goods and services = \$21,500

	American Ore, Inc.	American Steel, Inc.	American Motors, Inc.	Total factor income
Value of sales	\$4,200 (ore)	\$9,000 (steel)	\$21,500 (car)	
Intermediate goods	0	4,200 (iron ore)	9,000 (steel)	
Wages	2,000	3,700	10,000	\$15,700
Interest payments	1,000	600	1,000	2,600
Rent	200	300	500	1,000
Profit	1,000	200	1,000	2,200
Total expenditure by firm	4,200	9,000	21,500	
Value added per firm = Value of sales – Cost of intermediate goods	4,200	4,800	12,500	

3. Total payments to factors = \$21,500

1. Value of production of final goods and services, sum of value added = \$21,500

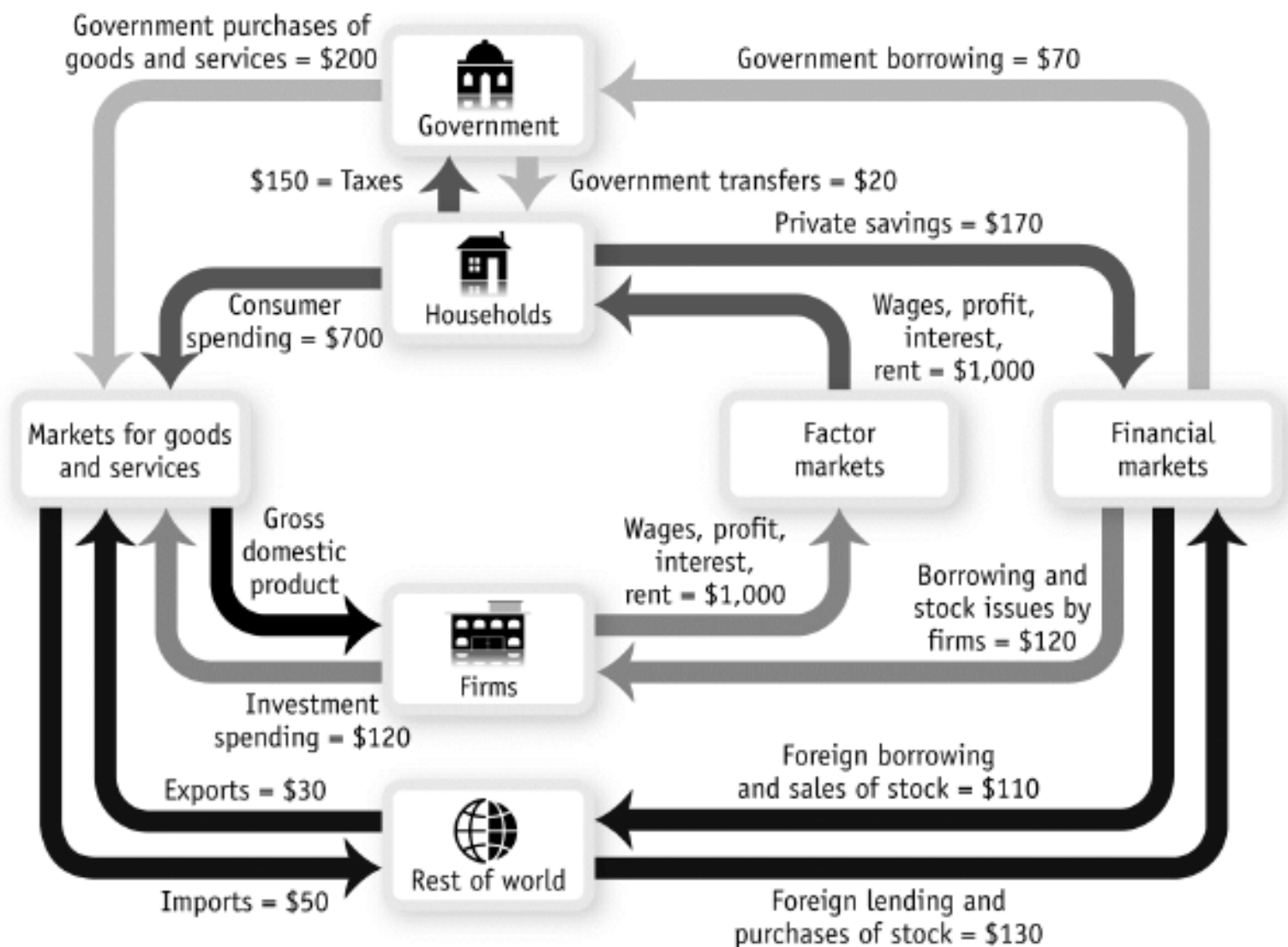


Table: Pizza Economy I

	Gino's pizza	Bruno's dough	Carlo's tomatoes	Aldo's pepperoni
Profit	\$4,000	\$3,000	\$2,000	\$5,000
Wages	10,000	9,000	6,000	7,000
Sales tax	4,000	1,000	1,000	2,000
Rent	8,000	6,000	1,000	4,000



LEARN BY DOING: PRACTICE QUESTION

Suppose country A sells \$100 million worth of goods and services to country B. Country B sells \$50 million worth of goods and services to country A. These are the only two countries in macro world. Net exports in country:

- a) B equal $-\$50$ million.**
- b) A equal $\$150$ million.**
- c) A equal $-\$150$ million.**
- d) B equal $\$50$ million.**

LEARN BY DOING: PRACTICE QUESTION

Income spent on imported goods:

- a) represents income that has leaked across national borders.**
- b) must be subtracted from spending data to calculate an accurate value for domestic production.**
- c) is income that is not spent on domestically produced goods and services.**
- d) Answers (a), (b), and (c) are all correct.**

ECONOMICS IN ACTION

WHO CREATED THE NATIONAL ACCOUNTS (AND WHY)?

The national accounts owe their creation to the Great Depression. The only data available were **scattered statistics** (railroad data, etc.)

Simon Kuznets (a young Russian-born economist) *developed a set of national income accounts. The first version was presented to Congress in 1937.*

The push to **complete the national accounts** came during **World War II**, when policy makers were **in even more need of comprehensive measures of the economy's performance.**

The federal government began issuing estimates of gross domestic product and gross national product in 1942.

REAL GDP: A MEASURE OF AGGREGATE OUTPUT

We need to be able track the quantity of total output over time.

Real GDP: the total value of the final goods and services produced in the economy during a given year, **calculated using the prices of a selected base year.**

Nominal GDP: the value of all final goods and services produced in the economy during a given year, **calculated using the prices current in the year in which the output is produced.**

REAL VS. NOMINAL GDP

Calculating GDP and real GDP in a simple economy:

how much would GDP have gone up if prices had *not* changed? To answer this question, we need to find the **value of output in year 2 expressed in year 1 prices.**

	Year 1	Year 2
Quantity of apples (billions)	2,000	2,200
Price of apple	\$0.25	\$0.30
Quantity of oranges (billions)	1,000	1,200
Price of orange	\$0.50	\$0.70
GDP (billions of dollars)	\$1,000	\$1,500
Real GDP (billions of year 1 dollars)	\$1,000	\$1,150

LEARN BY DOING: PRACTICE QUESTION

	Year 2010	Year 2011	Year 2012
Quantity of tractors	5	5	5
Price of tractors	\$10,000	\$12,000	\$15,000
Quantity of pizzas	100	80	100
Price of pizzas	\$10	\$20	\$20

Nominal GDP in 2011:

- a) is greater than nominal GDP in 2010.**
- b) is less than nominal GDP in 2012.**
- c) cannot be calculated until a base year is specified.**
- d) Answers (a) and (b) are both correct.**

LEARN BY DOING: PRACTICE QUESTION

	Year 2010	Year 2011	Year 2012
Quantity of tractors	5	5	5
Price of tractors	\$10,000	\$12,000	\$15,000
Quantity of pizzas	100	80	100
Price of pizzas	\$10	\$20	\$20

Suppose the base year is 2010. Real GDP in 2011 using 2010 as the base year:

- a) is greater than nominal GDP in 2011.**
- b) is less than nominal GDP in 2011.**
- c) is equal to nominal GDP in 2011.**
- d) may be greater than, less than, or equal to nominal GDP in 2011.**

LEARN BY DOING: PRACTICE QUESTION

	Year 2010	Year 2011	Year 2012
Quantity of tractors	5	5	5
Price of tractors	\$10,000	\$12,000	\$15,000
Quantity of pizzas	100	80	100
Price of pizzas	\$10	\$20	\$20

Using 2011 as the base year, real GDP is greatest in:

- a) 2010.**
- b) 2011.**
- c) 2012.**
- d) 2010 and 2012, since real GDP using 2011 as the base year is the same for these two years.**

REAL-LIFE REAL GDP

Nominal versus real GDP in 2005, 2009 and 2013

	Nominal GDP (billions of current dollars)	Real GDP (billions of 2009 dollars)
2005	\$13,094	\$14,234
2009	14,419	14,419
2013	16,768	15,710

REAL VS. NOMINAL GDP

Except in the base year, real GDP is not the same as **nominal GDP**: output valued at current prices.

Chained dollars: the method of calculating changes in real GDP using the average between the growth rate calculated on an early base year and the growth rate calculated on a late base year.

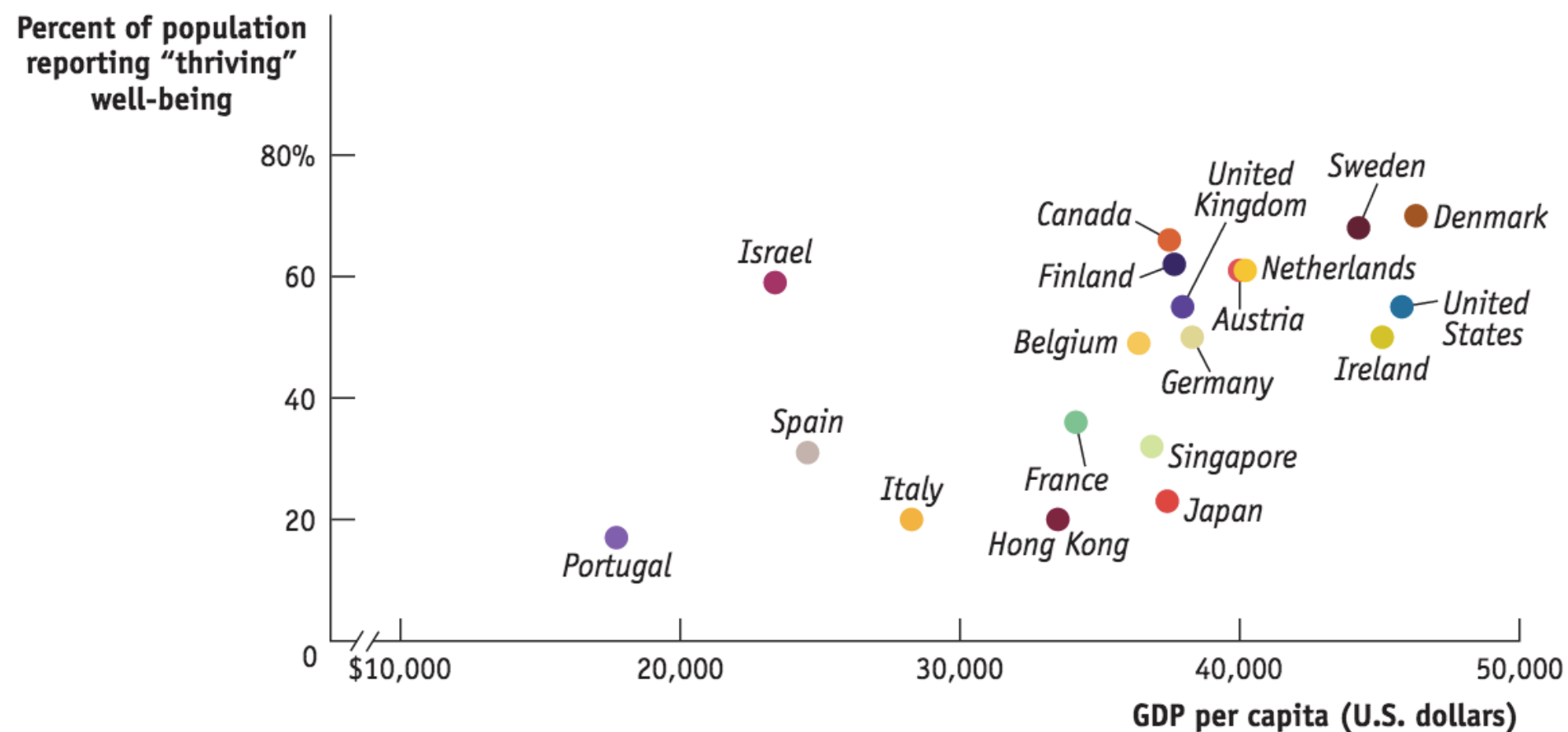
GDP per capita: average GDP per person; not by itself an appropriate policy goal.

GDP AND THE MEANING OF LIFE

Rich is better.

Money matters less as you grow richer.

Money isn't everything.

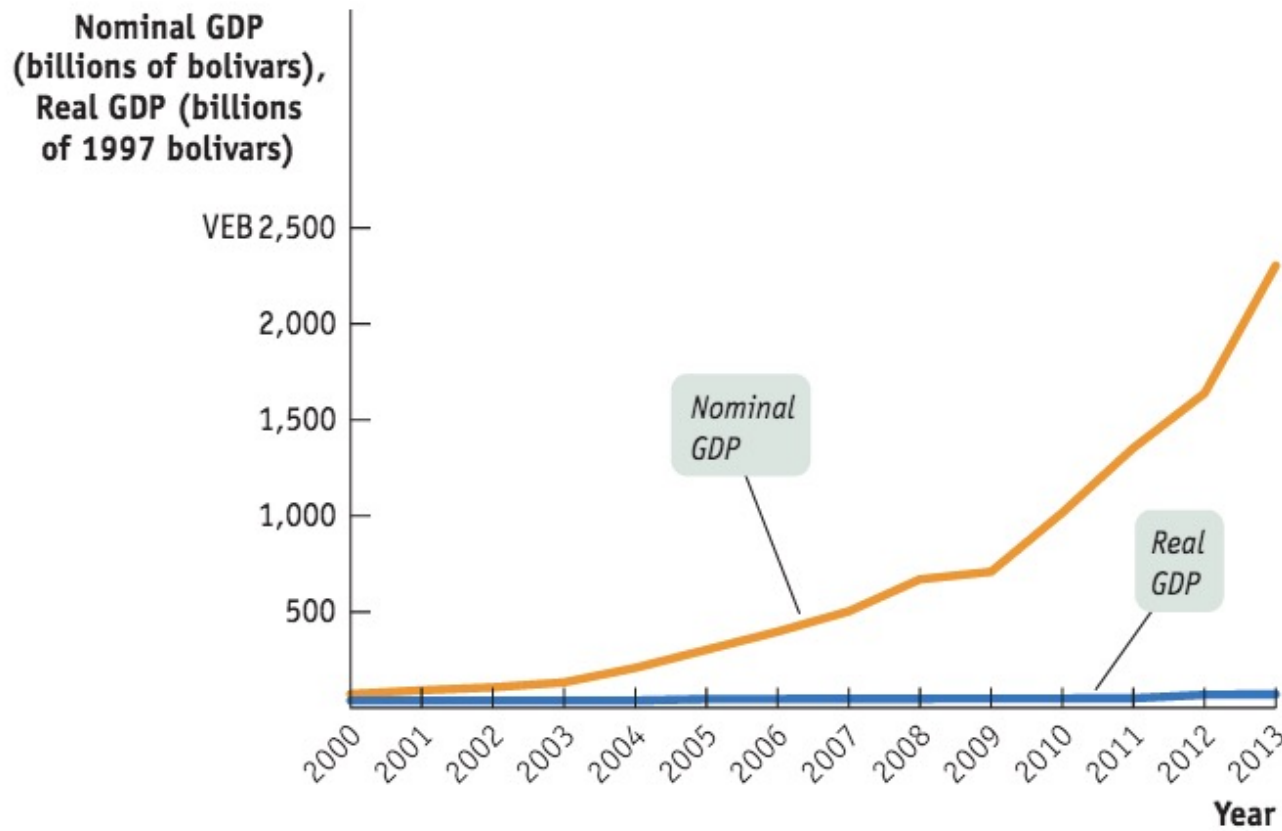


Source: Gallup; World Bank.

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MIRACLE IN VENEZUELA?

No, it's just suffering from unusually high inflation.



Source: IMF—World Economic Outlook (2014).

PRICE INDEXES AND THE AGGREGATE PRICE LEVEL

Aggregate price level: a measure of the overall level of prices in the economy.

To measure the aggregate price level, economists calculate the cost of purchasing a **market basket**.

Market basket: a hypothetical set of consumer purchases of goods and services.

MARKET BASKETS AND PRICE INDEXES

Calculating the cost of a market basket

	Pre-frost	Post-frost
Price of orange	\$0.20	\$0.40
Price of grapefruit	0.60	1.00
Price of lemon	0.25	0.45
Cost of market basket (200 oranges, 50 grapefruit, 100 lemons)	$(200 \times \$0.20) +$ $(50 \times \$0.60) +$ $(100 \times \$0.25) = \95.00	$(200 \times \$0.40) +$ $(50 \times \$1.00) +$ $(100 \times \$0.45) = \175.00

PRICE INDEXES

Price Index: the cost of purchasing a given market basket in a given year, where that cost is normalized so that it is equal to 100 in the selected base year

Price index in a given year

$$= \frac{\text{Cost of market basket in a given year}}{\text{Cost of market basket in base year}} \times 100$$

LEARN BY DOING: PRACTICE QUESTION

	Price in 2011	Price in 2012
Oranges	\$0.50	\$0.40
Apples	0.25	0.40
Bananas	0.40	0.50

Suppose 2011 is the base year and the market basket for purposes of constructing a price index consists of 200 oranges, 100 apples, and 100 bananas.

What is the value of the price index in 2011?

- a) 1.15**
- b) 100**
- c) 165**
- d) 190**

LEARN BY DOING: PRACTICE QUESTION

	Price in 2011	Price in 2012
Oranges	\$0.50	\$0.40
Apples	0.25	0.40
Bananas	0.40	0.50

What is the value of the price index in 2012, using 2011 as the base year?

- a) 0.97**
- b) 1**
- c) 100**
- d) 103**

INFLATION RATE, CPI, AND OTHER INDEXES

The ***Inflation rate***: the yearly percentage change in a price index, typically based upon ***consumer price index, or CPI***, the most common measure of the aggregate price level.

The ***consumer price index, or CPI***, measures the cost of the market basket of a typical urban American family

$$\text{Inflation rate} = \frac{\text{Price index in year 2} - \text{Price index in year 1}}{\text{Price index in year 1}} \times 100$$

LEARN BY DOING: PRACTICE QUESTION

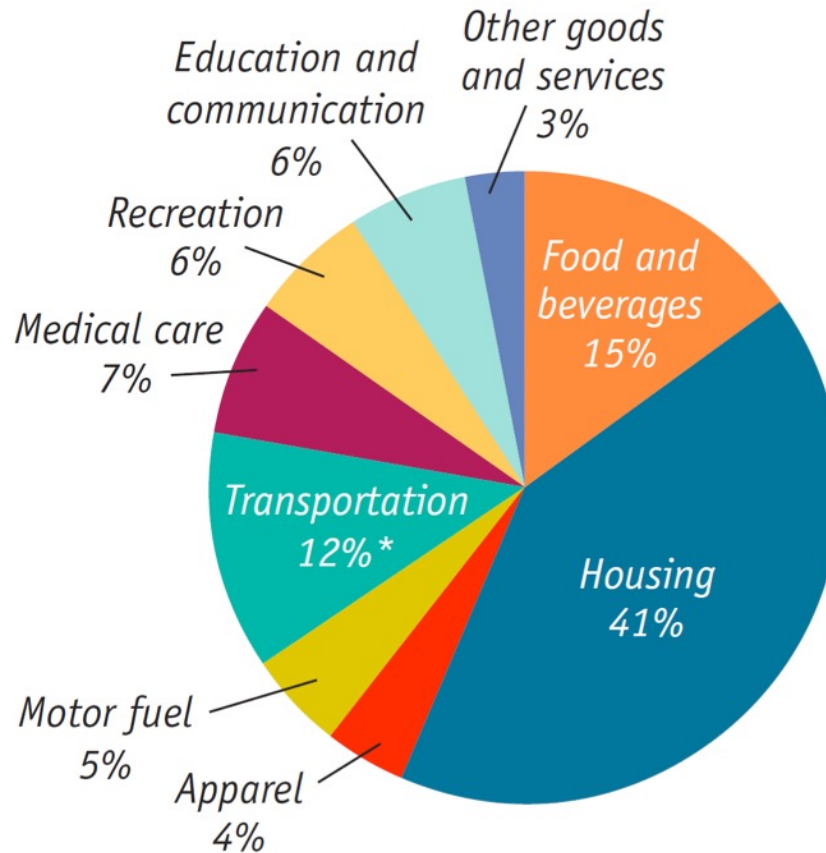
	Price in 2011	Price in 2012
Oranges	\$0.50	\$0.40
Apples	0.25	0.40
Bananas	0.40	0.50

What is the rate of inflation between 2011 and 2012 in this economy?

- a) 0%**
- b) 3%**
- c) 103%**
- d) 67%, since two of the three prices increased between 2011 and 2012**

THE CONSUMER PRICE INDEX

The makeup of the consumer price index in 2010

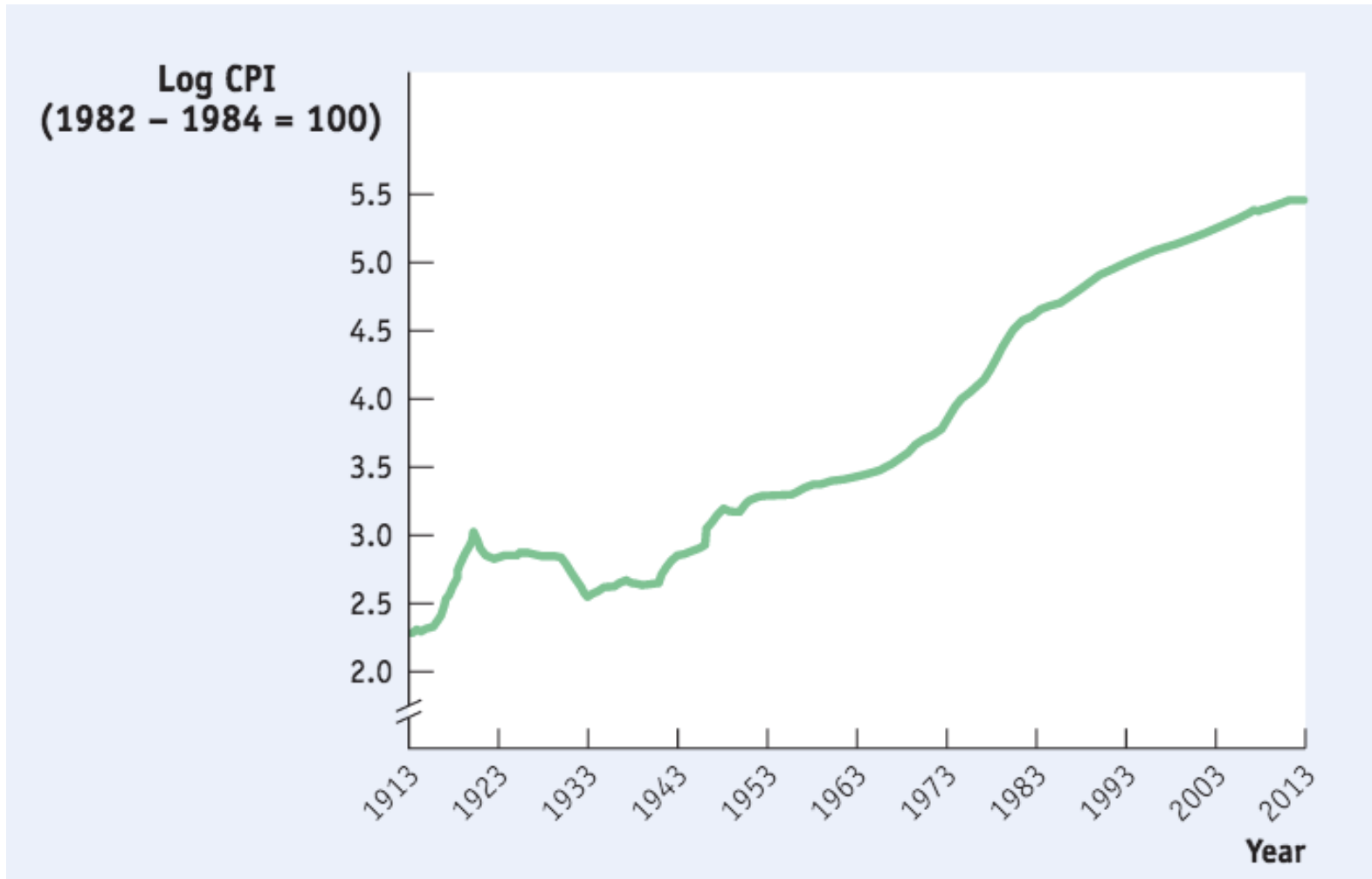


*Excludes motor fuel.

Source: Bureau of Labor Statistics.

CONSUMER PRICE INDEX

The annual percentage increases in recent years have been much smaller than those of the 1970s and early 1980s.



Source: Bureau of Labor Statistics.

OTHER PRICE MEASURES

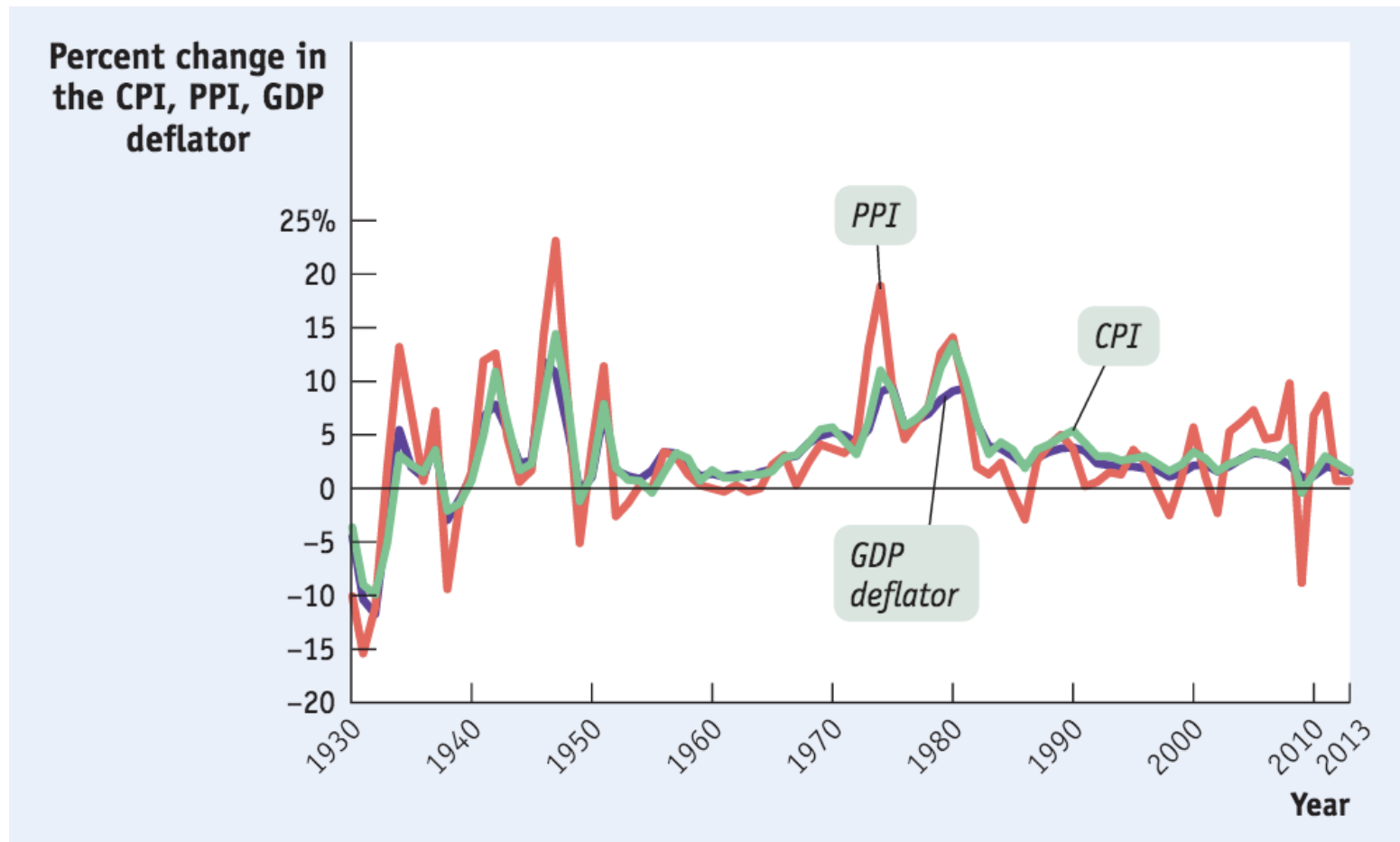
Producer price index (PPI): similar to the CPI but measures changes in the prices of goods purchased by producers.

Economists also use the ***GDP deflator***, which **measures the price level by calculating the ratio of nominal to real GDP.**

The GDP deflator for a given year is 100 times the ratio of nominal GDP to real GDP in that year.

THE CPI, THE PPI, AND THE GDP DEFLATOR

The three measures usually move closely together.



Source: Bureau of Labor Statistics.



INDEXING TO THE CPI

Many payments like Social Security are tied (“indexed”) to the CPI

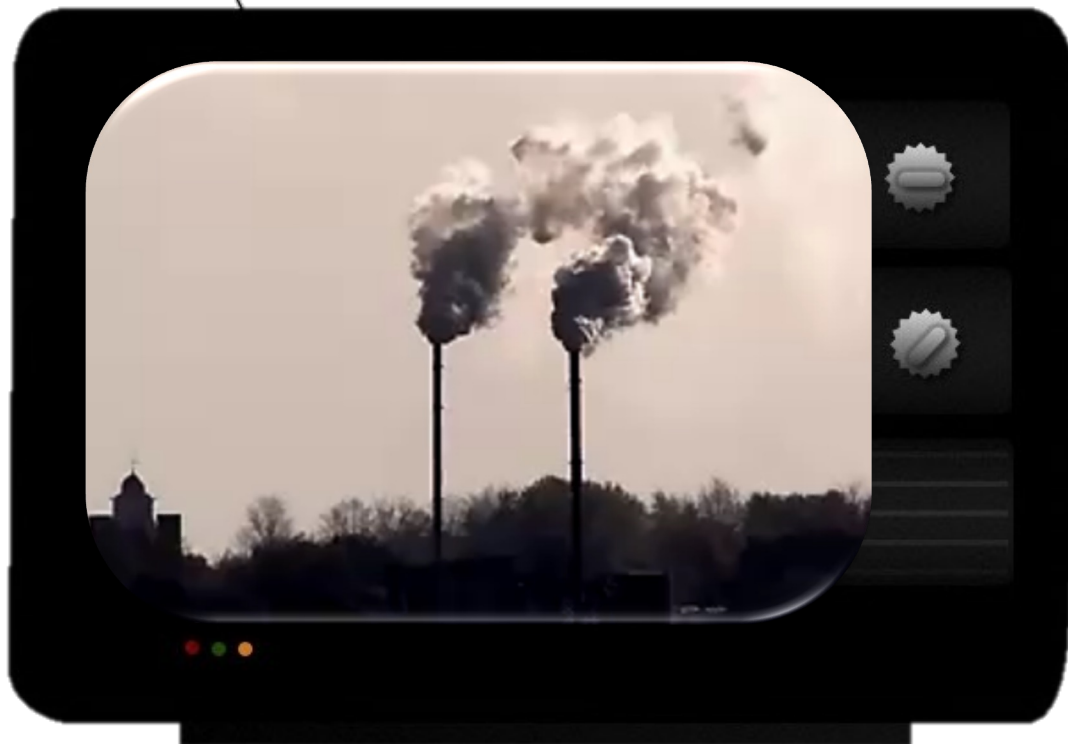
This practice goes back to the Revolutionary war: to keep soldiers’ pay in line with rising prices, wages were indexed to the cost of:

68 4/7 lbs beef, 10 lbs of wool and 16 lbs of leather



LEARN BY DOING: APPLICATION VIDEO

Does money buy happiness? Can well-being be measured? Does growth equal progress? These may sound like esoteric questions, but they are at the heart of efforts to move beyond GDP to develop more accurate ways of measuring human development touched on in [this](#) video. (3:19 minutes)



GETTING A JUMP ON GDP

Many businesses are eager for macroeconomic data and can't wait for the government's quarterly data

Two organizations supply monthly GDP estimates:

- **Macroeconomic Advisors** (based on raw government data)
- **Institute for Supply Management** (based on its own surveys of business orders)